

# [Foundational Clinic 3] Grade 4-5 Fraction Basics, Equivalent Fractions & Operations

Student Name: \_\_\_\_\_

Target: \_\_\_\_\_

Time: 35

Total: \_\_\_\_\_

Score: \_\_\_\_\_

🌸 Sophia & ⚡ William · Equivalent Fractions, Mixed Numbers, Division Reciprocal

Mins

100 Pts

## 💡 Key Fraction Laws (Root Error Correction):

① **Equivalent Fractions:** Whatever you multiply/divide the denominator by, you MUST do the exact same to the numerator!  $\frac{3}{4} = \frac{3 \times 5}{4 \times 5} = \frac{15}{20}$ . Never just multiply the denominator and forget the numerator!

② **Division with Remainder to Mixed Number:** In  $52 \div 24 = 2 \text{ R } 4$ , the quotient 2 is the whole number, the remainder 4 is the numerator, and 24 remains the denominator:  $2\frac{4}{24} = 2\frac{1}{6}$  (NEVER write  $24\frac{4}{24}$ !).

③ **Fraction Division:** Keep, Change, Flip!  $a \div \frac{b}{c} = a \times \frac{c}{b}$ . Always multiply by the reciprocal.

④ **Borrowing in Mixed Subtraction:** When  $3\frac{1}{5} - 1\frac{4}{5}$ , borrow 1 =  $\frac{5}{5}$  from 3:  $3\frac{1}{5} = 2\frac{6}{5}$ . Then  $2\frac{6}{5} - 1\frac{4}{5} = 1\frac{2}{5}$ .

## Section 1: Equivalent Fractions & Simplifying (3 pts each, 18 pts total)

Score: \_\_\_\_\_

Fill in the missing numerators or denominators to make the fractions equivalent, or simplify to lowest terms:

(a)  $\frac{3}{7} = \frac{\quad}{28}$  \_\_\_\_\_

(b)  $\frac{5}{8} = \frac{35}{\quad}$  \_\_\_\_\_

(c)  $\frac{18}{24} = \frac{\quad}{4}$  \_\_\_\_\_

(d) Simplify  $\frac{45}{60} =$  \_\_\_\_\_

(e) Simplify  $\frac{28}{42} =$  \_\_\_\_\_

(f)  $\frac{4}{9} = \frac{\quad}{36}$  \_\_\_\_\_

## Section 2: Improper Fractions & Mixed Numbers Conversion (3 pts each, 18 pts total)

Score: \_\_\_\_\_

Express as mixed numbers in simplest form (Quotient  $\frac{\text{Remainder}}{\text{Divisor}}$ ) or convert to improper fractions:

(a)  $52 \div 24 =$  \_\_\_\_\_

(b)  $\frac{38}{6} =$  \_\_\_\_\_

(c)  $\frac{47}{5} =$  \_\_\_\_\_

(d) Convert  $4\frac{3}{7}$  to improper fraction: \_\_\_\_\_

(e) Convert  $5\frac{2}{9}$  to improper fraction: \_\_\_\_\_

(f)  $75 \div 12 =$  \_\_\_\_\_

## Section 3: Fraction Addition & Subtraction (Common Denominator) (4 pts each, 24 pts total)

Score: \_\_\_\_\_

Show your common denominator calculation. Do NOT simply add across numerators and denominators!

(a)  $\frac{3}{4} + \frac{2}{5} =$   
LCD = 20. Expand BOTH numerators! \_\_\_\_\_

(b)  $\frac{5}{6} - \frac{3}{8} =$   
LCD = 24. \_\_\_\_\_

(c)  $2\frac{1}{3} + 1\frac{3}{5} =$   
Add whole numbers and fractions. \_\_\_\_\_

(d)  $3\frac{1}{4} - 1\frac{3}{4} =$   
Borrow 1 from 3 first! \_\_\_\_\_

(e)  $4\frac{2}{7} - 2\frac{5}{7} =$  \_\_\_\_\_

(f)  $\frac{7}{10} - \frac{1}{4} + \frac{2}{5} =$  \_\_\_\_\_

$$4\frac{2}{7} = 3\frac{9}{7}$$

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Find LCD of 10, 4, 5 (LCD = 20).

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**Section 4: Fraction Multiplication & Division (4 pts each, 20 pts total)**

Score: \_\_\_\_\_

Simplify before multiplying where possible. For division, multiply by the reciprocal (Keep-Change-Flip):

(a)  $\frac{4}{9} \times \frac{3}{8} =$  \_\_\_\_\_

(b)  $\frac{5}{12} \times 16 =$  \_\_\_\_\_

(c)  $\frac{7}{10} \div \frac{14}{15} =$  \_\_\_\_\_

(d)  $6 \div \frac{3}{4} =$  \_\_\_\_\_

(e)  $2\frac{1}{4} \times 1\frac{1}{3} \div \frac{9}{10} =$  (Convert to improper fractions first!) \_\_\_\_\_

**Section 5: Comparing Fractions & Word Applications (5 pts each, 20 pts total)** Score: \_\_\_\_\_1. [Comparing Fractions] Insert  $>$ ,  $<$ , or  $=$  in each blank:

(a)  $\frac{5}{12}$  \_\_\_\_\_  $\frac{7}{16}$

(b)  $\frac{11}{12}$  \_\_\_\_\_  $\frac{12}{13}$  (Residual method)

(c)  $\frac{7}{9}$  \_\_\_\_\_  $\frac{7}{11}$  (Same numerator)

## 2. [SUIS Practical Problem - Recipe Partition]

A baking recipe requires  $2\frac{1}{4}$  kg of flour. Sophia has a bag containing  $\frac{3}{5}$  kg of flour and William brings another bag containing  $1\frac{1}{2}$  kg of flour.

(a) What is the total mass of flour they have together?

(b) Do they have enough flour for the recipe? If not, how many kilograms of flour are they short of?

**Working & Verification:**

## 3. [Fraction of a Remaining Quantity]

A water tank has a capacity of 480 liters. On Monday,  $\frac{3}{8}$  of the water was used. On Tuesday,  $\frac{2}{5}$  of the **remaining** water was used. How many liters of water are left in the tank?**Step-by-step Solution:** **Complete Answer Key & Pedagogical Notes****Section 1 (Equivalent):**

- (a)  $12 (3 \times 4)$ ; (b)  $56 (8 \times 7)$ ; (c)  $3 (18 \div 6)$ ;  
(d)  $\frac{3}{4}$  (divide by 15); (e)  $\frac{2}{3}$  (divide by 14); (f)  $16 (4 \times 4)$ .

**Section 3 (Add/Subtract):**

- (a)  $\frac{15}{20} + \frac{8}{20} = \frac{23}{20} = 1\frac{3}{20}$ ;  
(b)  $\frac{20}{24} - \frac{9}{24} = \frac{11}{24}$ ;  
(c)  $2\frac{5}{15} + 1\frac{9}{15} = 3\frac{14}{15}$ ;  
(d)  $2\frac{3}{4} - 1\frac{3}{4} = 1\frac{2}{4} = 1\frac{1}{2}$  (borrow 1 as  $\frac{4}{4}$ );  
(e)  $3\frac{9}{7} - 2\frac{5}{7} = 1\frac{4}{7}$ ;  
(f)  $\frac{14}{20} - \frac{5}{20} + \frac{8}{20} = \frac{17}{20}$ .

**Section 2 (Conversions):**

- (a)  $2\frac{4}{24} = 2\frac{1}{6}$  (Sophia & William check: do not flip quotient & remainder!);  
(b)  $6\frac{2}{6} = 6\frac{1}{3}$ ; (c)  $9\frac{2}{5}$ ;  
(d)  $\frac{31}{7} (4 \times 7 + 3)$ ; (e)  $\frac{47}{9}$ ; (f)  $6\frac{3}{12} = 6\frac{1}{4}$ .

**Section 4 & 5 (Mult/Div & Applications):**

- Sec 4: (a)  $\frac{1}{6}$ ; (b)  $\frac{20}{3} = 6\frac{2}{3}$ ; (c)  $\frac{7}{10} \times \frac{15}{14} = \frac{3}{4}$ ; (d)  $6 \times \frac{4}{3} = 8$ ;  
(e)  $\frac{9}{4} \times \frac{4}{3} \times \frac{10}{9} = \frac{10}{3} = 3\frac{1}{3}$ .  
Sec 5: 1. (a)  $\frac{5}{12} = \frac{20}{48} < \frac{21}{48} = \frac{7}{16}$ ; (b)  $1 - \frac{1}{12} < 1 - \frac{1}{13} \implies \frac{11}{12} < \frac{12}{13}$ ; (c)  $\frac{7}{9} > \frac{7}{11}$ .  
2. (a)  $\frac{3}{5} + 1\frac{1}{2} = \frac{6}{10} + 1\frac{5}{10} = 2\frac{11}{10}$  kg; (b) No;  $2\frac{1}{4} - 2\frac{1}{10} = 2\frac{5}{20} - 2\frac{2}{20} = \frac{3}{20}$  kg short.  
3. Remaining after Mon:  $480 \times (1 - \frac{3}{8}) = 300$  L.

Tue used:  $300 \times \frac{2}{5} = 120$  L. Left:  $300 - 120 = 180$  L.